

✓ Practical 3

```
import csv
from collections import defaultdict
```

```
#MAP Function
def map_movies(file_path):
    mapped = []
    with open(file_path, 'r') as file:
        reader = csv.reader(file)
        next(reader)
        for row in reader:
            movie_id = row[0]
            rating = float(row[2])
            mapped.append((movie_id, [rating]))
    return mapped
```

```
# SHUFFLE & SORT FUNCTION
def shuffle_sort(mapped_data):
    grouped = defaultdict(list)
    for movie_id, rating_list in mapped_data:
        grouped[movie_id].extend(rating_list)
    return grouped
```

```
# REDUCE FUNCTION
def reduce_movies(grouped_data):
    reduced = {}
    for movie_id, ratings in grouped_data.items():
        avg = sum(ratings) / len(ratings)
        reduced[movie_id] = round(avg, 2)
    return reduced
```

```
import urllib.request
```

```
url = "https://raw.githubusercontent.com/zygmuntz/goodbooks-10k/master/ratings.csv"
urllib.request.urlretrieve(url, "ratings.csv")
```

```
('ratings.csv', <http.client.HTTPMessage at 0x7882ec137920>)
```

```
#Load CSV file
file_path = "ratings.csv"
mapped = map_movies(file_path)
grouped = shuffle_sort(mapped)
result = reduce_movies(grouped)
```

```
print("Average Ratings by Movies: ")

for movie_id, avg_rating in list(result.items())[:3]:
    print(f"Movie {movie_id}: {avg_rating}")
```

```
Average Ratings by Movies:
Movie 1: 3.59
Movie 2: 4.42
Movie 4: 3.77
```